PAUL HYUNBIN CHO

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U.S. Citizen

RESEARCH INTERESTS

My research focuses on video generation with diffusion models. I am interested in building video generative models that are controllable, efficient, and consistent over long horizons, spanning conditional video generation from signals such as audio, compact latent spaces for efficient generation, and autoregressive generation for streaming and long video synthesis. More broadly, I see video generation as a path toward world modeling, and I am interested in how video models can serve as interactive world simulators with persistent memory of the environments they generate.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Sep. 2025 - Present
M.S. in Artificial Intelligence Expected Aug. 2027
Supervisor: Seungryong Kim *Seoul, Korea*

Sogang University Feb. 2017 - Aug. 2023
B.A. in American Culture / B.S.E. in Computer Science and Engineering (Double Major) *Seoul, Korea*
Magna Cum Laude (GPA: 3.90/4.30)

PUBLICATIONS

(C: Conference, P: Preprint, J: Journal; *: equal contribution)

[C4] Geometry-Aware Representation Denoising for Robust Multi-view 3D Reconstruction
Jin Hyeon Kim*, Jaeeun Lee*, Claire Kim, Kyoungjin Oh, **Paul Hyunbin Cho**, Jaewon Min, Yeji Choi, Jihye Park, Hyunhee Park, Minkyu Park, Seungryong Kim
Neural Information Processing Systems (**NeurIPS**), 2026 [arXiv] [Project] [Code]

[C3] Deep Forcing: Training-Free Long Video Generation with Deep Sink and Participative Compression
Jung Yi, Wooseok Jang, **Paul Hyunbin Cho**, Jisu Nam, Heeji Yoon, Seungryong Kim
International Conference on Machine Learning (**ICML**), 2026 [arXiv] [Project] [Code]

[C2] Text-Aware Image Restoration with Diffusion Models
Jaewon Min, Jin Hyeon Kim, **Paul Hyunbin Cho**, Jaeeun Lee, Jihye Park, Minkyu Park, Sangpil Kim, Hyunhee Park, Seungryong Kim
International Conference on Learning Representations (**ICLR**), 2026 [arXiv] [Project] [Code]

[P5] GRACE: Generation-Aware Latent Compression for Efficient Video Generation
Jiyoung Kim, **Paul Hyunbin Cho**, Jisu Nam, Donghoon Lee, Hyunsung Go, Yeonkyeong Lee, Hansaem Kim, Seungryong Kim
arXiv preprint, 2026 (under review) [arXiv] [Project] [Code]

[P4] Lip Forcing: Few-Step Autoregressive Diffusion for Real-time Lip Synchronization
Paul Hyunbin Cho*, Jinhyuk Jang*, SeokYoung Lee, Joungbin Lee, Siyoon Jin, Heeseong Shin, Jung Yi, Yunjin Park, Chulmin Park, Seungryong Kim
arXiv preprint, 2026 (under review) [arXiv] [Project] [Code]

[P3] WorldKV: Efficient World Memory with World Retrieval and Compression
Jung Yi, Minjae Kim, **Paul Hyunbin Cho**, Wooseok Jang, Sangdoo Yun, Seungryong Kim
arXiv preprint, 2026 (under review) [arXiv] [Project] [Code]

[P2] DA-Flow: Degradation-Aware Optical Flow Estimation with Diffusion Models
Jaewon Min, Jaeeun Lee, Yeji Choi, **Paul Hyunbin Cho**, Jin Hyeon Kim, Tae-Young Lee, Jongsik Ahn, Hwayeong Lee, Seonghyun Park, Seungryong Kim
arXiv preprint, 2026 (under review) [arXiv] [Project] [Code]

[P1] UniT: Unified Diffusion Transformer for High-fidelity Text-Aware Image Restoration
Jin Hyeon Kim, **Paul Hyunbin Cho**, Claire Kim, Jaewon Min, Jaeeun Lee, Jihye Park, Yeji Choi, Seungryong Kim
arXiv preprint, 2025 [arXiv] [Project] [Code]

[C1] Enhancing Sidewalk Maintenance through Accurate Joint Deflection Measurement
Jisoo Park*, Seonghak Lee*, **Paul Hyunbin Cho***, Zongkai Xia, Loucif Hebbache
IEEE International Conference on Consumer Electronics (**ICCE**), 2025 [DOI]

[J1] Efficient GNN-based Social Recommender Systems through Social Graph Refinement
Sangmin Ga, **Paul Hyunbin Cho**, Gordon Euhyun Moon, Sungwon Jung
The Journal of Supercomputing, 2024 [DOI]

EXPERIENCE

Kakao Corp. Mar. 2026 - Present
Research Intern *Seongnam, Korea*
Project 1: Acceleration of Audio-driven Portrait Animation with Autoregressive Video Diffusion Models
- Distilled an audio-driven portrait animation model into a few-step autoregressive video diffusion model for streaming generation.

Project 2: Latent Compression for Efficient Video Diffusion [P5]
- Co-designed and evaluated GRACE, which compresses the latent space of a pretrained video autoencoder to 8× fewer tokens while keeping DiT compatibility; further step-distilled it into bidirectional and autoregressive few-step models, for a 117.7× total speedup over Wan2.1-I2V-14B.

AIPARK Aug. 2025 - Dec. 2025
Research Intern *Seoul, Korea* (Remote)
Project: Real-time Audio-driven Lip Synchronization with Autoregressive Video Diffusion Models [P4]
- Led Lip Forcing: distilled a 14B bidirectional teacher into causal autoregressive students via Sync-Window DMD and a SyncNet-based reward; the 1.3B student runs in real time at 31 FPS with two-step inference.

KAIST CVLAB Sep. 2024 - Apr. 2025
Research Intern *Seoul, Korea*
Project: Text-Aware Image Restoration with Diffusion Models [C2]

University of Toronto Dec. 2023 - Jul. 2024
Visiting Graduate Researcher *Toronto, Canada*
Project: Stereo Vision-based Sidewalk Defect Detection and Measurement [C1]

Ratio Co. Ltd. Sep. 2023 - Jul. 2024
Graduate Research Assistant *Seoul, Korea*
Project: GNN-based Bicycle Route Recommendation System [J1]

ADDITIONAL INFORMATION

Citizenships: United States, Republic of Korea
Languages: Native proficiency in both English and Korean
Programming: Python, PyTorch, CUDA, C/C++